

APSPDCL · COMMERCIAL ROOFTOP

800 KWP GRID-TIED SOLAR PLANT

HT CATEGORY I / II AT 11 KV

30-YEAR ANALYSIS

Net investment ₹2.90 Cr after input tax credit, generating 12,26,400 units in year one against an energy charge of ₹7.65 per unit.

PAYBACK 3.5 yrs Post-tax, undiscounted	PROJECT IRR 29.0% Unlevered, 30 years	COST OF SOLAR ENERGY ₹2.98 per kWh, against ₹7.65 from the grid	NET PRESENT VALUE ₹5.85 Cr Discounted at 10%
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🏠 PLANT AND GENERATION

Plant capacity	800 kWp
Year 1 generation	12,26,400 kWh
Specific yield	1,533 kWh / kWp / year
Supply category	HT Category I / II at 11 kV (HT supply at 11 kV)
Energy charge offset	₹7.65 / kWh
Units self-consumed	90% at the energy charge
Units exported	10% at ₹2.75 / kWh (APPC)
Year 1 gross saving	₹87.81 L

₹ INVESTMENT AND TAX BASIS

Price per watt-peak	₹38.00 / Wp (incl. GST)
Project price incl. GST	₹3.04 Cr
GST at 5% — recovered as ITC	₹14.48 L
Net capitalised cost	₹2.90 Cr
Depreciation block	40% WDV, Appendix I
Tax regime	s.115BAA — 25.168%
Annual maintenance	₹600 / kWp / year, year 1 free
30-year depreciation shield	₹72.87 L

📈 ACCELERATED DEPRECIATION — FIRST 12 YEARS

YEAR	OPENING WDV (₹)	DEPRECIATION AT 40% (₹)	CLOSING WDV (₹)	TAX SHIELD AT 25.168% (₹)
1	2,89,52,381	1,15,80,952	1,73,71,429	29,14,694
2	1,73,71,428	69,48,571	1,04,22,857	17,48,816
3	1,04,22,857	41,69,143	62,53,714	10,49,290
4	62,53,715	25,01,486	37,52,229	6,29,574
5	37,52,228	15,00,891	22,51,337	3,77,744
6	22,51,337	9,00,535	13,50,802	2,26,647
7	13,50,802	5,40,321	8,10,481	1,35,988
8	8,10,482	3,24,193	4,86,289	81,593
9	4,86,289	1,94,516	2,91,773	48,956
10	2,91,773	1,16,709	1,75,064	29,373
11	1,75,064	70,026	1,05,038	17,624
12	1,05,038	42,015	63,023	10,574
Tax shield, years 1-12				72,70,874

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

30-year post-tax cash flow

800 kWp
PLANT CAPACITY

How this table works. Electricity savings reduce an operating expense, so they increase taxable profit and are taxed at 25.168%. Depreciation reduces the same taxable profit. Both are applied to one figure, so the tax column is a net inflow in the early years when depreciation exceeds savings — shown in green. Net cash flow is therefore gross saving, less maintenance, less that tax. The shaded row is the year the investment is recovered.

YEAR	GENERATION (KWH)	GROSS SAVING (₹)	AMC (₹)	DEPRECIATION (₹)	TAX (₹)	NET CASH FLOW (₹)	CUMULATIVE (₹)
1	12,26,400	87,81,024	—	1,15,80,952	(7,04,686)	94,85,710	94,85,710
2	12,20,268	90,86,604	4,80,000	69,48,571	4,17,294	81,89,310	1,76,75,020
3	12,14,167	94,02,817	4,80,000	41,69,143	11,96,405	77,26,413	2,54,01,433
4	12,08,096	97,30,035	4,80,000	25,01,486	16,98,475	75,51,560	3,29,52,993
5	12,02,055	1,00,68,641	4,80,000	15,00,891	20,35,525	75,53,116	4,05,06,109
6	11,96,045	1,04,19,029	4,80,000	9,00,535	22,74,808	76,64,221	4,81,70,330
7	11,90,065	1,07,81,612	4,80,000	5,40,321	24,56,722	78,44,890	5,60,15,220
8	11,84,115	1,11,56,812	4,80,000	3,24,193	26,05,547	80,71,265	6,40,86,485
9	11,78,194	1,15,45,069	4,80,000	1,94,516	27,35,901	83,29,168	7,24,15,653
10	11,72,303	1,19,46,837	4,80,000	1,16,709	28,56,600	86,10,237	8,10,25,890
11	11,66,441	1,23,62,587	4,80,000	70,026	29,72,985	89,09,602	8,99,35,492
12	11,60,609	1,27,92,805	4,80,000	42,015	30,88,312	92,24,493	9,91,59,985
13	11,54,806	1,32,37,995	4,80,000	25,209	32,04,587	95,53,407	10,87,13,392
14	11,49,032	1,36,98,677	4,80,000	15,126	33,23,070	98,95,607	11,86,08,999
15	11,43,287	1,41,75,391	4,80,000	9,075	34,44,572	1,02,50,819	12,88,59,818
16	11,37,571	1,46,68,695	4,80,000	5,445	35,69,640	1,06,19,054	13,94,78,872
17	11,31,883	1,51,79,165	4,80,000	3,267	36,98,664	1,10,00,501	15,04,79,373
18	11,26,223	1,57,07,400	4,80,000	1,960	38,31,939	1,13,95,461	16,18,74,834
19	11,20,592	1,62,54,018	4,80,000	1,176	39,69,709	1,18,04,309	17,36,79,143
20	11,14,989	1,68,19,657	4,80,000	706	41,12,187	1,22,27,470	18,59,06,613
21	11,09,414	1,74,04,981	4,80,000	423	42,59,573	1,26,65,409	19,85,72,022
22	11,03,867	1,80,10,675	4,80,000	254	44,12,056	1,31,18,619	21,16,90,641
23	10,98,348	1,86,37,446	4,80,000	152	45,69,828	1,35,87,619	22,52,78,260
24	10,92,856	1,92,86,029	4,80,000	91	47,33,078	1,40,72,951	23,93,51,211
25	10,87,392	1,99,57,183	4,80,000	55	49,02,004	1,45,75,180	25,39,26,391
26	10,81,955	2,06,51,693	4,80,000	33	50,76,803	1,50,94,890	26,90,21,281
27	10,76,545	2,13,70,372	4,80,000	20	52,57,684	1,56,32,688	28,46,53,969
28	10,71,162	2,21,14,061	4,80,000	12	54,44,858	1,61,89,204	30,08,43,173
29	10,65,807	2,28,83,630	4,80,000	7	56,38,544	1,67,65,087	31,76,08,260
30	10,60,478	2,36,79,981	4,80,000	4	58,38,970	1,73,61,011	33,49,69,271
Total	3,42,44,965	45,18,10,921	1,39,20,000	2,89,52,373	10,29,21,654	33,49,69,271	33,49,69,271

COMMERCIAL SOLAR — FINANCIAL ANALYSIS

Return metrics, assumptions and basis

800 kWp
PLANT CAPACITY

RETURN METRICS

Simple payback	3.5 yrs
Discounted payback	4.5 yrs
Project IRR, unlevered	29.0%
Net present value at 10%	₹5.85 Cr

OVER THE FULL TERM

Levelised cost of energy	₹2.98 / kWh
Cumulative return, 30 years	1057%
Total net cash flow, after tax	₹33.50 Cr
30-year generation	3,42,44,965 kWh

COST OF A UNIT — SOLAR AGAINST THE GRID

Solar, levelised over 30 years		₹2.98
Grid energy charge today		₹7.65

Every unit the plant generates costs ₹2.98 across its life against ₹7.65 bought from APSPDCL today — and the grid figure reaches ₹23.86 by year 30 at 4% escalation, while the solar figure is fixed at the day of purchase. Over 30 years the plant generates 3,42,44,965 units, avoiding 28,080.9 tonnes of CO₂.

ASSUMPTIONS

- Tariff:** HT Category I / II at 11 kV at ₹7.65/kWh. Energy charge; ₹475/kVA/month demand charge excluded. APERC Retail Supply Tariff Order FY2025-26, continued for FY2026-27.
- Demand charges:** Excluded. Solar does not reduce contracted demand, so fixed and demand charges are unchanged by the plant.
- Tariff escalation:** 4% per year, applied to both the energy charge and the export rate.
- Generation:** 1,533 kWh per kWp in year one, degrading 0.5% per year.
- Unit split:** 90% self-consumed, 10% exported at ₹2.75/kWh. The plant is assumed to be sized to the site's daytime load.
- GST:** 5%, recovered in full as input tax credit. Figures are stated ex-GST. A buyer who cannot claim ITC should read the cost as ₹3.04 Cr.
- Depreciation:** 40% written-down value, Income Tax Act Appendix I. No additional depreciation under s.32(1)(ia), which s.115BAA forfeits.
- Tax:** 25.168% — s.115BAA at 22% plus 10% surcharge and 4% cess. Applied to savings and to depreciation alike.
- Loss set-off:** Early-year depreciation exceeds savings. The resulting loss is assumed to shelter the buyer's other business income in the same year.
- Commissioning:** Full first-year depreciation assumed, i.e. the plant is put to use on or before 30 September. Later commissioning halves the year-one charge, with the balance carried forward.
- Maintenance:** ₹600 per kWp per year from year 2, held flat. Year 1 is covered by the build.
- Inverter replacement:** Not modelled. A mid-life inverter replacement would reduce the figures shown.
- Financing:** None. All returns are unlevered — the return on the money invested, before any loan.
- Horizon:** 30 years, which runs beyond the module performance warranty.
- Discount rate:** 10% for net present value, discounted payback and levelised cost.
- Emissions:** Grid emission factor 0.82 kg CO₂ per kWh; tree equivalence at 45 trees per tonne is illustrative, not measured sequestration.

BASIS AND LIMITATIONS

This is an indicative financial analysis for a standard 800 kWp commercial rooftop plant. **It is not a quotation, an offer, or tax advice.** Pricing is subject to a site survey, roof condition, structure height, cable routing, evacuation distance and the APSPDCL sanction actually granted.

Tariffs, the depreciation block and the corporate tax rate are as we understand them at the date printed below and are subject to change by APERC and by the Finance Act. The tax treatment shown assumes a GST-registered company that has opted into s.115BAA and has sufficient other business income to absorb the early-year depreciation. **Have your chartered accountant confirm the tax position before you rely on it.**

Generation depends on site irradiance, shading, orientation, soiling and grid availability. Actual results will differ from this projection.